

Direct-yaw-moment control of four-wheel-drive electrical vehicle based on lateral tyre-road forces and sideslip angle observer

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Abstract: Considering some technical and economic reasons, it is not easy to directly measure the vehicular moving parameters (such as tyre-road forces and vehicle sideslip angle) in electronic stability programme systems. This study proposes a method to estimate lateral tyre-road forces and vehicle sideslip angle by utilising real-time measurements, based on the unscented Kalman filter. Direct-yaw-moment control can effectively guarantee the stability of vehicle while steering at a high speed. This study proposed a hierarchical control strategy as the solution to the problem of the yaw-moment distribution. The overloop controller is designed to calculate the desired yaw moment based on the estimated lateral tyre-road forces and sideslip angle, using the sliding mode control. The servo-loop controller is designed to optimise the torque distribution using weighted-least-squares method based on the desired yaw moment obtained from the overloop controller. MATLAB/Simulink with Carsim is applied for the simulation experiment, the results demonstrate the effectiveness of the lateral tyre-road force and sideslip angle observer, and the optimal allocation controller could improve the handling stability and energy efficiency dramatically.

1 Introduction

Electric vehicle (EV) as a kind of clean-energy vehicle has recently become flourished; moreover, lots of automotive companies and universities are developing EV. The EV is not only energy efficient and environment friendly but also good from a control point of view. Comparing to EV, the conventional vehicles which have drive shafts and differentials limit the range of power distribution/differential torque capability and have a longer response time. The EV equipped with in-wheel motors has superior performance due to the accurate torque can be generated by a quick response of the motor during deceleration and acceleration. The wheel torque can be calculated easily from the electric current value of in-wheel motor and the driving wheel torque can be controlled easily. However, the EV's security has always been a concern for the people, especially the handling stability of EV. The dangerous conditions including 'sideslip', 'flick' and 'sharp turns' are more likely to occur which causes potential traffic accidents when the EV is fast on the low road coefficient. To avoid the occurrence of potential traffic accidents, a growing number of researchers have been devoted to the research on the stability control for EV. At present, the well known vehicle stability control methods mainly include the direct-yaw-moment control, antilock braking system and traction control system, and they have been widely used in the vehicle [1, 2]. The literatures [3–13] proposed the control strategies based on the direct-yaw-control (DYC). The yaw rate, sideslip angle, longitudinal and lateral tyre-road forces *etc.* are the important variables which reflect or affect the vehicle state. However, the researches above ignore the important state vectors of the vehicle, especially the lateral tyre-road forces of wheels. Therefore, this paper proposed a hierarchical control strategy as the solution to the problem of the yaw-moment distribution for four-wheel-drive (4WD) electrical vehicle based on the lateral tyre-road forces and sideslip angle observer.

2 Literature review

Recently, DYC is a basic approach to vehicle lateral motion control, and it could effectively improve vehicle lateral stability through differential traction/braking [1, 2].

In recent years, a number of studies on DYC have been published. The paper [14] proposed an energy-efficient control allocation scheme to distribute control efforts for over-actuated systems by explicitly incorporating efficiency functions and working modes of redundant actuators; the study [3] proposed an in-wheel-motor torque distribution method for the electric ground vehicle to show the potentials of optimising the four independently actuated in-wheel motors electric ground vehicle operational energy efficiency; De Novellis *et al.* [4] assessed the performance of alternative objective functions for the optimal wheel torque distribution of a four-wheel-drive fully EV with a newly developed offline optimisation procedure; the research [15] proposed a novel fuzzy sliding mode control (SMC) method to solve destabilisation because of high non-linear characteristics and parameter uncertainty of distributed drive EV when turning or changing lanes at high speeds; the papers [16, 17] studied four motors torque distribution problem to improve its energy-saving capacity and efficiency, by constructing an objective function which minimised the vehicle output power to optimise the driving or braking torque. The literatures [18, 19] proposed wheel torques vector-control algorithm which reduced the wheel slip rate effectively and improved the stability of the vehicle; Saikia and Mahanta [5] designed two-level controller, the SMC methodology is used in the upper-level control in order to generate the corrective steering wheel angle and yaw moment, the yaw moment is realised through braking between appropriate wheels in the second-level control; Ding *et al.* [6] proposed the direct-yaw-moment control strategies for in-wheel EVs by using SMC and non-linear disturbance observer techniques; the research [7] proposed a motor braking-based DYC strategy, and this strategy could improve the stability of vehicle; the paper [8] proposed a vehicle dynamics control system to compensate the change in-vehicle handling dynamics of light-weight vehicles due to variation in loading conditions; the study [9] proposed a novel DYC method-emergency control algorithm and adaptive control algorithm based on driver operation intention for stability control of a distributed drive EV; the paper [10] proposed an upper yaw-moment controller and a lower-torque distribution controller based on direct-yaw-moment control; Chen *et al.* [11] studied relationship between the vehicle longitudinal velocity and lateral stability, and proposed a novel DYC system

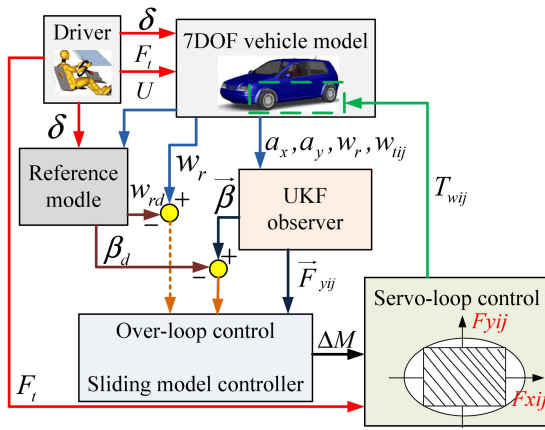


Fig. 1 Scheme of integrated vehicle chassis controller

with lighter control effort and better control effect; Zhang *et al.* [12] proposed a DYC strategy for in-wheel EVs by using second-order SMC technique, and first constructed a conventional discontinuous sliding mode DYC controller to guarantee that the sideslip angle and yaw rate will approach the ideal ones as closely as possible; the literature [13] proposed a new sliding mode DYC control scheme with two adaptive controller parameters, in order to enhance the system robustness against uncertainties and disturbances; the paper [20] proposed a new optimal strategy for differential braking to recover the vehicle stability in emergency manoeuvres, an optimal non-linear control law is analytically developed for calculating the external yaw moment and the distributed braking forces in the upper layer, a non-linear wheel slip tracking controller is designed for the front wheels to generate the required braking forces in the lower layer; the study [21] designed an H_∞ controller to determine yaw rate to restrain the effect of uncertainties such as parameter perturbation and lateral wind disturbance.

As mentioned above, most of the DYC controllers are designed, supposing the lateral tyre-road forces and sideslip angle have been accurately measured or calculated, especially the lateral tyre road. Actually, the lateral tyre-road forces and sideslip angle are difficult to be measured, and the cost is very expensive.

Since the control system of vehicle motion is mostly generated from tyre-road contact patches, the accuracy of tyre force generation is definitely a key factor that determines whether the intention of driver or chassis controller could be realised. The proposed controller needs to know the vehicle responses, the yaw rate, the sideslip angle and tyre-road force in order to generate the suitable output. The yaw rate can be directly measured by a yaw rate sensor (gyroscope) or can be estimated with an observer. In general, the sideslip angle and tyre-road force have to be estimated using an observer.

Shi *et al.* [22] first obtained the heading angle of the vehicle through the global positioning system, and then took the heading angle and yaw rate as correction amount of the vehicle side angle observer, improving the estimation precision of sideslip angle; Rajamani [23] estimated sideslip angle based on a linear two-degree of freedom (2DOF) vehicle model; Jin and Yin [24] estimated the lateral force of vehicle tyres and sideslip angle using an interactive multi-model filtering method; the study [25] proposed a method which first combines auxiliary particle filter and the iterated extended Kalman filter (EKF), and then processed the results of the first step using iteration algorithm to estimate the slip angle of the front tyres with strong non-linearity and non-Gaussian noise; the research [26] proposed an individual tyre force estimation algorithm that is particularly designed for all-wheel-drive vehicles using the interacting multiple model filter method; Yoon *et al.* [27] presented a method that estimates the vehicle sideslip angle and a tyre-road friction coefficient by combining measurements of a magnetometer, a global positioning system and an inertial measurement unit; the paper [28] designed an observer for the sideslip angle of ground vehicles based on the uncertain linear-parameter-varying system; the literature [29] proposed a novel method to estimate lateral tyre-road forces and vehicle

sideslip angle by utilising real-time measurements based on an interacting multiple model filter that integrated in-vehicle sensors of in-wheel-motor-driven EVs to adapt multiple vehicle-road system models to variable driving conditions; the research [30] designed a non-linear observer for sideslip angle of four-in-wheel-motor-driven EVs with the estimated information on tyre cornering stiffness.

The aim of this paper is to improve the stability of EV based on the DYC, considering the problem that longitudinal forces and sideslip angle could not be measured directly. However, the methods mentioned above supposed that the lateral forces and sideslip angle could be measured accurately, especially the lateral forces. Therefore, this paper begins with the estimated observer of longitudinal forces and sideslip angle based on the unscented Kalman filter (UKF). The second part of this paper conducts a robust DYC controller and an optimal distribution strategy based on the estimated longitudinal forces and sideslip angle, improving the accuracy of controllers. This paper is organised as follows. In Section 3, the scheme of integrated vehicle chassis controller is proposed, In Section 4, the vehicle and the tyre models are briefly presented. The adopted vehicle model is a 7DOF model which includes the non-linearity that is presented in the vehicle system. The non-linear tyre model considered is the magic formula tyre model. In Section 5, the UKF observer of lateral tyre-road forces and sideslip angle is designed. In Section 6, the desired yaw-moment controller is proposed based on the estimated lateral tyre-road forces and sideslip, aiming at improving vehicle handling stability. Additionally, an optimal distribute strategy is proposed based on the desired yaw moment. In Section 7, the results stemmed from the simulation in Simulink/Carsim for the different driving condition are shown and discussed. Finally, conclusions are presented in Section 8.

3 Scheme of integrated controller

The scheme of integrated vehicle chassis controller is shown as Fig. 1.

The scheme of integrated vehicle chassis controller consists of three parts based on the inputs of the driver: the UKF observer, the overloop controller and the servo-loop controller:

- In the UKF observer section, the sideslip angle and tyre-road forces are estimated based on the measurable variables (longitudinal acceleration, lateral acceleration, yaw rate and the angular velocity of tyres), aiming at to ensure the accuracy of desired yaw moment, which is calculated by the sideslip angle and lateral tyre-road forces.
- In the overloop, in order to track the desired yaw rate and sideslip angle of the reference model, a sliding mode controller based on the yaw rate and sideslip angle errors is designed to obtain the desired yaw moment M_d .
- In the servo-loop, the desired yaw moment would be optimally allocated to tyre actuator actions. The friction circle limitation at each wheel needs to be considered, consisting of the tyre-road adhesion limits.

4 Vehicle and tyre models

4.1 2DOF vehicle model

The 2DOF vehicle model is employed here to calculate the ideal sideslip angle and yaw rate. The model is based on the assumption that the left and right tyre slip angles are equal, and the roll motion is ignored. Moreover, the lateral force generated by the tyre is proportional to the tyre slip angle. Therefore, the vehicle dynamical model with 2DOF only consists of lateral motion and yaw motion. The 2DOF vehicle model is shown as Fig. 2. Then, the vehicle motion equations are given as follows:

$$m\dot{a}_y = 2k_f(\beta + aw_r/U - \delta) + 2k_r(\beta - bw_r/U) \quad (1)$$

$$I_z \dot{w}_r = 2ak_f(\beta + aw_r/U - \delta) - 2bk_r(\beta - bw_r/U) \quad (2)$$

where $a_y = \dot{v} + U w_r$, $\beta = v/U$, a_y is the lateral acceleration, β is the sideslip angle, m is total mass of the vehicle, v is the lateral velocity, U is the longitudinal velocity, w_r is the yaw rate, k_f and k_r are the lateral stiffnesses, respectively, a is the distance between the centre of mass and the front axle, b is the distance between the centre of mass and the rear axle, δ is the steering angle of front wheel and I_z is the yaw inertia moment.

The maximum value of the yaw rate is given in [31]

$$w_{r_u} = \mu g/U \quad (3)$$

where w_{r_u} is the maximum value of the yaw rate and μ is the road friction coefficient.

4.2 7DOF vehicle model

Fig. 3 shows the 7DOF vehicle model used in this paper. The coordinate system of the vehicle meets the right-handed system, and the positive direction of the X-axis is the forward direction of the vehicle. The 7DOF vehicle model includes longitudinal motion, lateral motion, yawing motion and the rotational dynamics of four wheels. The equations of motion for the 7DOF vehicle model are derived from Fig. 3:

Longitudinal motion

$$m(\dot{U} - v w_r) = (F_{xfl} + F_{xfr}) \cos \delta + F_{xrl} + F_{xrr} - (F_{yfl} + F_{yfr}) \sin \delta \quad (4)$$

Lateral motion

$$m(\dot{v} + U w_r) = (F_{xfl} + F_{xfr}) \sin \delta + F_{yrl} + F_{yrr} + (F_{yfl} + F_{yfr}) \cos \delta \quad (5)$$

The sideslip angle can be expressed as

$$\beta = v/U \quad (6)$$

Substituting (6) into (5), lateral motion can be expressed as (see (7)). Yaw motion can be expressed as

$$I_z \dot{w}_r = a(F_{xfl} + F_{xfr}) \sin \delta + a(F_{yfr} + F_{yfl}) \cos \delta + \Delta M_z - b(F_{yrl} + F_{yrr}) + \frac{t_f}{2}(F_{yfl} - F_{yfr}) \sin \delta \quad (8)$$

where $\Delta M_z = (t_f/2)(F_{xfr} - F_{xfl}) \cos \delta + (t_f/2)(F_{xrr} - F_{xrl})$ being the yaw-moment controller.

Wheels rotational motion

$$I_{wi} \dot{w}_{ti} = -T_i - R F_{xi} \quad (9)$$

where m is the total mass of the vehicle, U is the longitudinal velocity, v is the lateral velocity, w_r is the yaw rate, β is the sideslip angle, F_{xij} and F_{yij} are longitudinal and lateral tyre-road forces, respectively. $ij = fl, fr, rl, rr$; δ is the steering angle of the front wheel, a is the distance between the centre of the mass and the front axle, b is the distance between the centre of the mass and the rear axle, t_f is the track width, I_z is the yaw inertia moment, I_{wi} is the moment of inertia of the wheel, w_{ti} is the angular velocity, T_i is wheel torque, $i = fl, fr, rl, rr$ and R is the radius of the wheel.

This model includes a quasi-static lateral and longitudinal load transfer. Thus, the normal load equation for each wheel can be expressed as

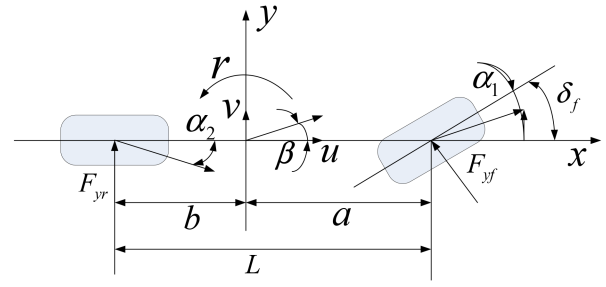


Fig. 2 2DOF vehicle model

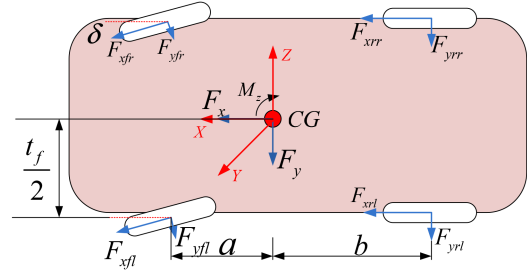


Fig. 3 7DOF vehicle model

$$F_{zfl} = mg \frac{b}{2L} - ma_x \frac{h}{2L} - ma_y \frac{h}{L t_f} + ma_x a_y \frac{h^2}{g L t_f} \quad (10)$$

$$F_{zfr} = mg \frac{b}{2L} - ma_x \frac{h}{2L} + ma_y \frac{h}{L t_f} - ma_x a_y \frac{h^2}{g L t_f} \quad (11)$$

$$F_{zrl} = mg \frac{a}{2L} + ma_x \frac{h}{2L} - ma_y \frac{h}{L t_f} + ma_x a_y \frac{h^2}{g L t_f} \quad (12)$$

$$F_{zrr} = mg \frac{a}{2L} + ma_x \frac{h}{2L} + ma_y \frac{h}{L t_f} - ma_x a_y \frac{h^2}{g L t_f} \quad (13)$$

where $L = a + b$.

Moreover, the longitudinal and lateral accelerations can be expressed as

$$a_x = \frac{1}{m}(F_{xfl} + F_{xfr}) \cos \delta + \frac{1}{m}(F_{xrl} + F_{xrr}) - \frac{1}{m}(F_{yfl} + F_{yfr}) \sin \delta \quad (14)$$

$$a_y = \frac{1}{m}(F_{xfl} + F_{xfr}) \sin \delta + \frac{1}{m}(F_{yrl} + F_{yrr}) + \frac{1}{m}(F_{yfl} + F_{yfr}) \cos \delta \quad (15)$$

4.3 Tyre model

Lateral and longitudinal tyre-road forces are a function of the tyre properties and the other factors of tyre. The relation between these factors is extremely complex and non-linear [32, 33]. To calculate the force, a tyre-road force model is required. The Magic Formula tyre model is one of the most well known tyre model [34]. On the basis of the Magic Formula tyre model, the longitudinal and lateral forces can be calculated by the equation as follows:

$$\begin{cases} \bar{F}_x = D_x \sin\{C_x \arctan[B_x \kappa - E_x(B_x \kappa - \arctan(B_x \kappa))]\} \\ \bar{F}_y = D_y \sin\{C_y \arctan[B_y \alpha - E_y(B_y \alpha - \arctan(B_y \alpha))]\} \end{cases} \quad (16)$$

$$\begin{aligned} mU \dot{\beta} &= (F_{xfl} + F_{xfr})(\sin \delta - \beta \cos \delta) + F_{yrl} + F_{yrr} - mU w_r \\ &\quad - m\beta^2 U w_r + (F_{yfl} + F_{yfr})(\cos \delta + \beta \sin \delta) - \beta(F_{xrl} + F_{xrr}) \end{aligned} \quad (7)$$

The pure longitudinal and lateral tyre–road force can be calculated with (16). The longitudinal and lateral tyre–road forces at combined slip are given as (17)

$$\begin{cases} F_x = -\frac{\sigma_x}{\sqrt{\sigma_x^2 + \sigma_y^2}} \bar{F}_x \\ F_y = \frac{\sigma_y}{\sqrt{\sigma_x^2 + \sigma_y^2}} \bar{F}_y \end{cases} \quad (17)$$

According to the 7DOF vehicle, each wheel has an independent slip angle

$$\begin{cases} \alpha_{fl} = \delta - \arctan\left((v + aw_r)/\left(U - \frac{tf}{2}w_r\right)\right) \\ \alpha_{fr} = \delta - \arctan\left((v + aw_r)/\left(U + \frac{tf}{2}w_r\right)\right) \end{cases} \quad (18)$$

$$\begin{cases} \alpha_{rl} = \arctan\left((-v + bw_r)/\left(U - \frac{tf}{2}w_r\right)\right) \\ \alpha_{rr} = \arctan\left((-v + bw_r)/\left(U + \frac{tf}{2}w_r\right)\right) \end{cases} \quad (19)$$

Moreover, the longitudinal wheel slip is defined as

$$\kappa_i = \begin{cases} (U - R w_i)/U, & U \geq R w_i \\ (R w_i - U)/R w_i, & R w_i > U \end{cases} \quad (20)$$

5 UKF observer design

5.1 Lateral tyre–road force and sideslip angle observer

The Kalman filter observer mainly focuses on the application of EKF. However, EKF brings in errors while transferring non-linear functions into linear ones by Taylor expansion, which makes parameter estimation hard to achieve expected accuracy. Compared to EKF, UKF has the following advantages: (i) it adopts the non-linear probability density of the non-linear function to approximate, rather than the non-linear function; (ii) the computational accuracy of non-linear statistics reaches order 2 at least; (iii) it need not differentiate Jacobian matrix; (iv) it can deal with non-additive noise and discrete systems; and (v) it uses deterministic sampling strategy to avoid the problem of particle decline [35].

In this section, for the estimation of sideslip angle and lateral tyre–road forces, an observer is designed based on the UKF algorithm with the symmetry rule sampling strategy. Fig. 4 shows the principle of UKF for each single model filter.

For the UKF observer design, the transition and measurement models are expressed as follows:

$$\begin{aligned} \mathbf{x}_{k+1} &= f(\mathbf{x}_k, \mathbf{u}_k, t_k) + \mathbf{w}_k \\ y_k &= h(\mathbf{x}_k, \mathbf{u}_k, t_k) + v_k \end{aligned} \quad (21)$$

where $\mathbf{x}$ is a state vector, $\mathbf{u}$ is the input, y is measurement, k is the index for time steps and w_k and v_k are the process and measurement noises, respectively, which are assumed to be zero-mean Gaussian noise with covariance Q_k and R_k , respectively. The design synthesis of the UKF is described as follows [36]:

i. Initialisation

$$\begin{cases} \hat{\mathbf{x}}(0|0) = E[\mathbf{x}(0)] \\ P_{xx}(0|0) = E[(\mathbf{x}(0) - \hat{\mathbf{x}}(0|0)) \cdot (\mathbf{x}(0) - \hat{\mathbf{x}}(0|0))^T] \end{cases} \quad (22)$$

ii. Sigma point calculation and time update

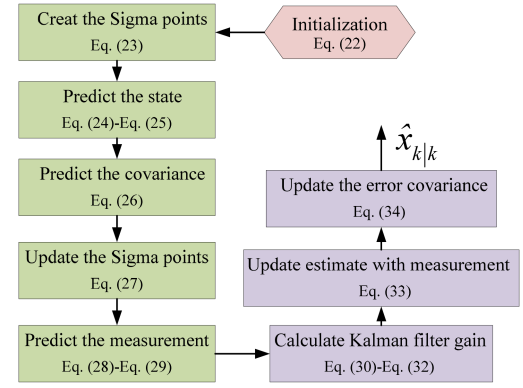


Fig. 4 Principle of UKF for each single model filter

$$\sigma_i(k|k) = \begin{bmatrix} \hat{\mathbf{x}}(k|k), \hat{\mathbf{x}}(k|k) + \sqrt{(n+\lambda)P_{xx}(k|k)}, \\ \hat{\mathbf{x}}(k|k) - \sqrt{(n+\lambda)P_{xx}(k|k)} \end{bmatrix}^T \quad (23)$$

$$\sigma_i(k+1|k) = f(\sigma_i(k|k), u(k)) \quad (24)$$

$$\hat{\mathbf{x}}(k+1|k) = \sum_{i=0}^{2n} W_i^m \sigma_i(k+1|k) \quad (25)$$

$$P_{xx}(k+1|k) = \sum_{i=0}^{2n} W_i^c [\sigma_i(k+1|k) - \hat{\mathbf{x}}(k+1|k)] \times [\sigma_i(k+1|k) - \hat{\mathbf{x}}(k+1|k)]^T + Q$$

$$\begin{aligned} \hat{\sigma}_i(k+1|k) &= \begin{bmatrix} \hat{\mathbf{x}}(k+1|k), \hat{\mathbf{x}}(k+1|k) + \sqrt{(n+\lambda)P_{xx}(k+1|k)}, \\ \hat{\mathbf{x}}(k+1|k) - \sqrt{(n+\lambda)P_{xx}(k+1|k)} \end{bmatrix}^T \end{aligned} \quad (27)$$

$$\hat{Y}_i(k+1|k) = h(\hat{\sigma}_i(k+1|k)) \quad (28)$$

$$\hat{y}_i(k+1|k) = \sum_{i=0}^{2n} W_i^m \hat{Y}_i(k+1|k) \quad (29)$$

iii. Measurement update

$$P_{yy}(k+1|k) = \sum_{i=0}^{2n} W_i^c [\hat{Y}_i(k+1|k) - \hat{y}_i(k+1|k)] \times [\hat{Y}_i(k+1|k) - \hat{y}_i(k+1|k)]^T + R \quad (30)$$

$$P_{xy}(k+1|k) = \sum_{i=0}^{2n} W_i^c [\hat{\sigma}_i(k+1|k) - \hat{\mathbf{x}}(k+1|k)] \times [\hat{Y}_i(k+1|k) - \hat{y}_i(k+1|k)]^T \quad (31)$$

$$K(k+1) = P_{xy}(k+1|k)P_{yy}^{-1}(k+1|k) \quad (32)$$

$$\hat{\mathbf{x}}(k+1|k+1) = K(k+1)[y(k+1) - \hat{y}(k+1|k)] + \hat{\mathbf{x}}(k+1|k) \quad (33)$$

(see (34))

where Q and R are processed noise covariance and measurement noise covariance, respectively.

$$\begin{aligned} P_{xx}(k+1|k+1) &= -K(k+1)P_{yy}(k+1|k)K(k+1)^T \\ &\quad + P_{xx}(k+1|k) \end{aligned} \quad (34)$$

In this paper, the longitudinal and lateral tyre–road force should be added to the state vector. Therefore, the state vector is defined as

$$\mathbf{x} = \begin{bmatrix} U, \beta, w_r, w_{tfl}, w_{tfr}, w_{trl}, w_{trr}, F_{xfl}, F_{xfr}, \\ F_{xrl}, F_{xrr}, F_{yfl}, F_{yfr}, F_{yrl}, F_{yrr} \end{bmatrix}^T \quad (35)$$

The measurement vector and estimable variables are defined as

$$\mathbf{z} = [a_x, a_y, w_r, w_{tfl}, w_{tfr}, w_{trl}, w_{trr}]^T \quad (36)$$

$$\mathbf{x}_{\text{estimated}} = [\beta, F_{yfl}, F_{yfr}, F_{yrl}, F_{yrr}]^T$$

The input vector is defined as

$$\mathbf{u} = [\delta, T_{fl}, T_{fr}, T_{rl}, T_{rr}]^T \quad (37)$$

The particular non-linear function $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ of the state is derived as follows:

$$\dot{U} = \frac{1}{m}((F_{xfl} + F_{xfr})\cos\delta + F_{xrl} + F_{xrr} - (F_{yfl} + F_{yfr})\sin\delta) + v w_r \quad (38)$$

$$\dot{\beta} = \frac{1}{mU} \left((F_{xfl} + F_{xfr})(\sin\delta - \beta \cos\delta) + F_{yrl} + F_{yrr} - m\beta^2 U w_t \right) \quad (39)$$

$$\left((F_{yfl} + F_{yfr})(\cos\delta + \beta \sin\delta) - \beta(F_{xrl} + F_{xrr}) - mU w_r \right)$$

$$\dot{w}_r = \frac{1}{I_z} \left(a(F_{xfl} + F_{xfr})\sin\delta + \frac{t_f}{2}(F_{yfr} - F_{yfl})\cos\delta \right. \quad (40)$$

$$\left. + \frac{t_f}{2}(F_{xrr} - F_{xrl}) - b(F_{yrl} + F_{yrr}) + \Delta M_z \right)$$

$$\dot{w}_{tfl} = (-T_{fl} - \text{RF}_{xfl})/I_{wfl} \quad (41)$$

$$\dot{w}_{tfr} = (-T_{fr} - \text{RF}_{xfr})/I_{wfr} \quad (42)$$

$$\dot{w}_{trl} = (-T_{rl} - \text{RF}_{xrl})/I_{wrl} \quad (43)$$

$$\dot{w}_{trr} = (-T_{rr} - \text{RF}_{xrr})/I_{wrr} \quad (44)$$

$$\dot{F}_{xfl} = 0 \quad (45)$$

$$\dot{F}_{xfr} = 0 \quad (46)$$

$$\dot{F}_{xrl} = 0 \quad (47)$$

$$\dot{F}_{xrr} = 0 \quad (48)$$

The particular observation $z = h(\mathbf{x}, \mathbf{u})$ is derived as follows:

$$U = U \quad (49)$$

$$a_x = \frac{1}{m}(F_{xfl} + F_{xfr})\cos\delta + \frac{1}{m}(F_{xrl} + F_{xrr}) - \frac{1}{m}(F_{yfl} + F_{yfr})\sin\delta \quad (50)$$

$$a_y = \frac{1}{m}(F_{xfl} + F_{xfr})\sin\delta + \frac{1}{m}(F_{yrl} + F_{yrr}) + \frac{1}{m}(F_{yfl} + F_{yfr})\cos\delta \quad (51)$$

$$w_r = w_r \quad (52)$$

$$w_{tfl} = w_{tfl} \quad (53)$$

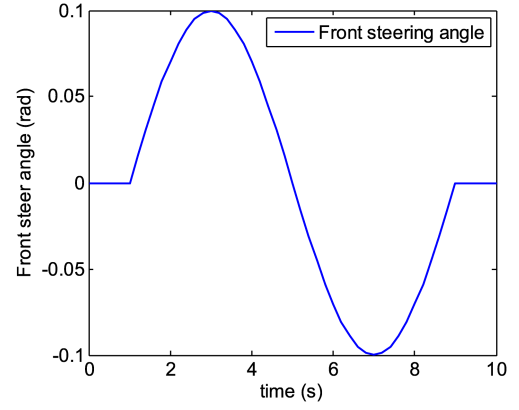


Fig. 5 Input steering angle

$$w_{tfr} = w_{tfr} \quad (54)$$

$$w_{trl} = w_{trl} \quad (55)$$

$$w_{trr} = w_{trr} \quad (56)$$

Using equations mentioned above, applying the UKF can estimate the state vector $\mathbf{x}$.

5.2 Simulation and analysis

To testify the effectiveness of the proposed state observer, this paper compared the simulation results related to different road friction coefficient conditions (1 and 0.7). Assuming the initial speed of the vehicle is 33 km/h and driving follows the sinusoidal steering angle input, the steering angle input is shown as Fig. 5 and the simulation results are shown as Figs. 6 and 7, respectively.

As shown in Figs. 6 and 7, the lateral tyre–road force and sideslip angle estimated by the proposed method are almost equal to the actual ones under different road friction coefficient conditions. Therefore, the UKF observer is effective to estimate the lateral tyre–road force and sideslip angle and is able to provide the exact value for the following controllers.

6 Controllers design

This paper designs a hierarchical controller based on the estimated lateral tyre–road forces and sideslip angle, and the structure of the control system is shown as Fig. 8. The upper-loop controller is designed to calculate the desired yaw moment based on the estimated lateral tyre–road forces and sideslip angle, using the SMC. The servo-loop controller is designed to optimise the torque distribution based on the desired yaw moment obtained from the overloop controller, using the weighted-least-squares (WLS) method.

6.1 Overloop controller design

SMC is a well known non-linear control method for dealing with uncertain non-linear systems. In this section, the upper controller is designed based on SMC. First of all, we construct a controller by using the conventional first-order SMC method. Then, we further use the second-order SMC to eliminate the chattering problem existing in the first-order SM controller.

The first thing to design an SM controller is to choose a sliding surface. Note that the actual sideslip angle and the yaw rate should approach their ideal value. Hence, the sliding surface of SMC is defined as

$$s = w_r - w_{rd} + \lambda(\beta_e - \beta_d) \quad (57)$$

where w_r and w_{rd} are actual yaw rate and ideal yaw rate, respectively. β_e and β_d are estimated sideslip angle and desired sideslip angle, respectively. λ is coefficient and $\lambda > 0$.

Differentiating (57) gives the following equation:

$$\dot{s} = \dot{w}_r - \dot{w}_{rd} + \lambda(\dot{\beta}_e - \dot{\beta}_d) \quad (58)$$

Substituting (8) into (58), the equation can be expressed as follows:

$$\begin{aligned} \dot{s} = & \frac{1}{I_z}(a(F_{xfl} + F_{xfr})\sin\delta + a(F_{yfr} + F_{yfl})\cos\delta \\ & -b(F_{yrl} + F_{yrr}) + \frac{t_f}{2}(F_{yfl} - F_{yfr})\sin\delta + \Delta M_z) \\ & -\dot{w}_{rd} + \lambda(\dot{\beta}_e - \dot{\beta}_d) \end{aligned} \quad (59)$$

where F_{xi} and F_{yi} are the estimated longitudinal force and lateral force of wheels, respectively, $i = fl, fr, rl, rr$.

The equivalent control input that would achieve $\dot{s} = 0$ is (see (60)). To satisfy sliding condition regardless of the uncertainty and disturbances on system dynamics, a discontinuous term is added to ΔM_{z_deal}

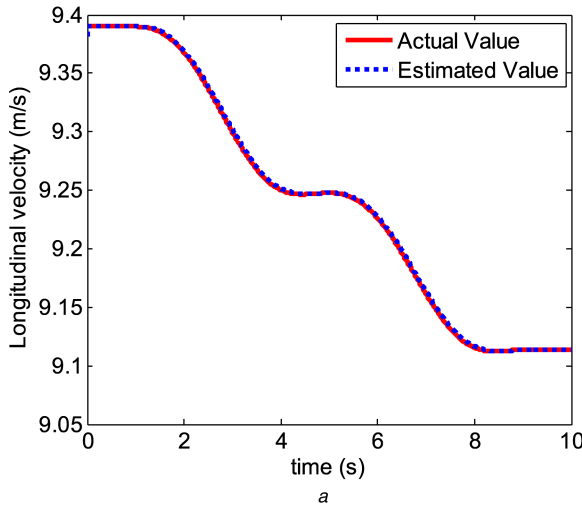
$$\Delta M_{z_deal} = \Delta M_z - k_1 s - k_2 \text{sign}(s) \quad (61)$$

where k_1 should be large enough, $k_2 > 0$.

To eliminate the chatters due to the high-frequency component of the control input, a saturation function $\text{sat}(s/p)$ is used as a continuous approximation of $\text{sign}(s)$. The desired yaw-moment control input is represented as follows:

$$\Delta M_{z_deal} = \Delta M_z - k_1 s - k_2 \text{sat}(s/p) \quad (62)$$

$$\text{sat}(x) = \begin{cases} x, & \text{if } |x| \leq 1 \\ \text{sign}(x), & \text{if } |x| > 1 \end{cases} \quad (63)$$



6.2 Servo-loop controller design

The DYC law has been discussed in the preceding section. However, since it is infeasible to apply yaw moment to motor vehicle on the road from an external source, a study was conducted on a technique for controlling yaw moment through proper distribution of traction and braking forces between the wheels.

In this section, the optimal distribute strategy is proposed based on the desired yaw moment, aiming to distribute the desired yaw moment to the wheels.

However, the calculated final control inputs using this approach should satisfy the below condition [37]:

Issue 1: Minimise the allocation error of the desired yaw moment.

Issue 2: Maximisation of the energy efficiency driving.

Issue 3: Actuator constraints.

Issue 1: Minimise the allocation error. The overlevel control inputs and the actuator commands are related as follows:

$$\begin{cases} F_x = F_{xfl} + F_{xfr} + F_{xrl} + F_{xrr} \\ \Delta M = \frac{t_f}{2}(-F_{xfl} + F_{xfr} - F_{xrl} + F_{xrr}) \end{cases} \quad (64)$$

The upper-level control inputs V are defined as

$$V = [F_x, \Delta M]^T \quad (65)$$

The actuator commands are defined as

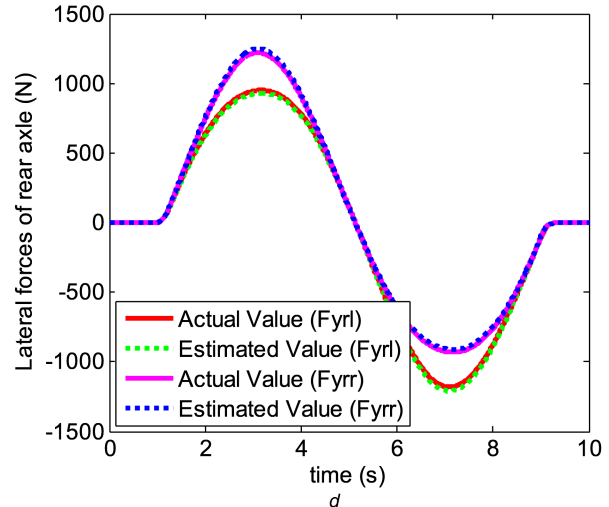
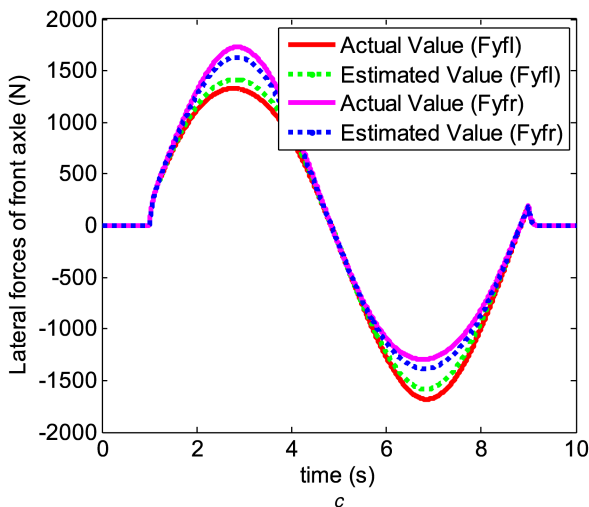
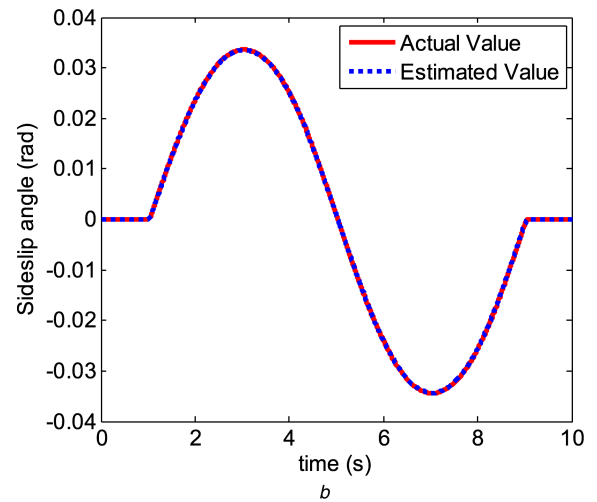


Fig. 6 Comparison of the simulation results (road friction coefficient is 1)

(a) Longitudinal velocity, (b) Sideslip angle, (c) Lateral forces of front tyres, (d) Lateral forces of rear tyres

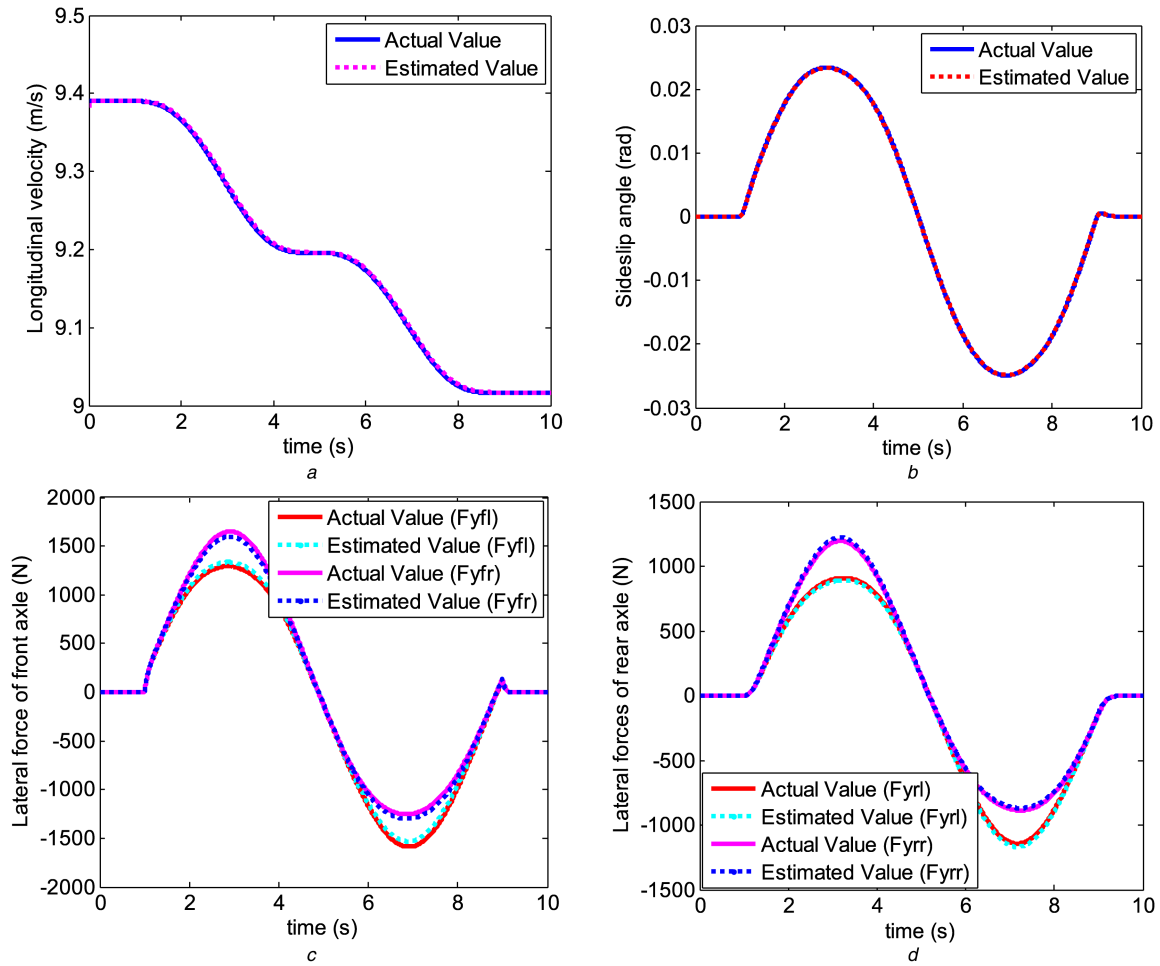


Fig. 7 Comparison of the simulation results (road friction coefficient is 0.7)
(a) Longitudinal velocity, (b) Sideslip angle, (c) Lateral forces of front tyres, (d) Lateral forces of rear tyres

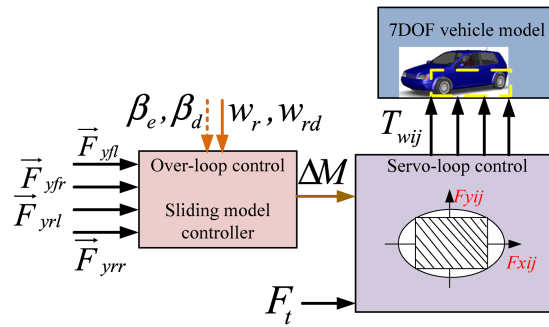


Fig. 8 Structure of control system

$$\Delta M_z = -\frac{t_f}{2}(F_{yfl} - F_{yfr})\sin \delta + I_z(\dot{w}_{rd} - \lambda(\dot{\beta}_e - \dot{\beta}_d)) - a[(F_{xfl} + F_{xfr})\sin \delta - (F_{yfr} + F_{yfl})\cos \delta] + b(F_{yrl} + F_{yrr}) \quad (60)$$

$$\mathbf{u} = [F_{xfl}, F_{xfr}, F_{xrl}, F_{xrr}]^T \quad (66)$$

Equation (64) can be expressed as follows:

$$\mathbf{V} = \mathbf{B}\mathbf{u} \quad (67)$$

where $\mathbf{B}$ is the efficiency matrix

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{t_f}{2} & \frac{t_f}{2} & -\frac{t_f}{2} & \frac{t_f}{2} \end{bmatrix} \quad (68)$$

The control allocation problem is an under-determined and often constrained problem. A common approach to solving these problems is to formulate an optimisation problem, in which the allocation error

$$u_i(t) = \operatorname{argmin} \| \mathbf{W}_v(\mathbf{B}\mathbf{u} - \mathbf{V}) \|^2 \quad (69)$$

The 2-norm of the allocation error is most commonly used [9]. In (69), the weighting matrix for the control allocation error is as follows:

$$\mathbf{W}_v = \operatorname{diag}([1 \quad 10]) \quad (70)$$

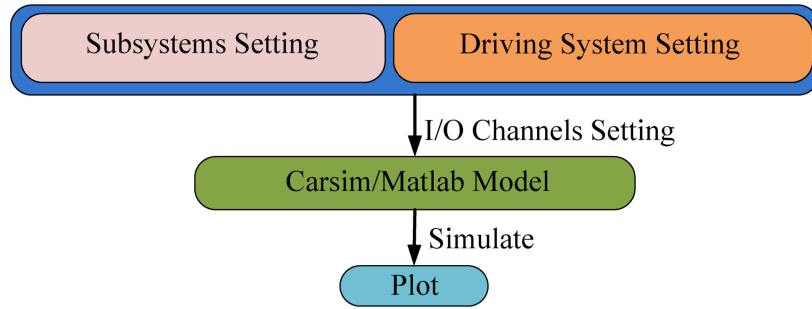


Fig. 9 Simulation platform of Simulink/Carsim

The demand of yaw moment is a priority when the demand of longitudinal force cannot be satisfied, which has certain practical significance for the lateral stability of the vehicle.

Issue 2: Energy efficiency driving: The energy efficiency driving can be described as follows:

$$u_2(t) = \arg\min \|W_u u\|^2 \quad (71)$$

In (71), the weighting matrix for the efficient driving is as follows:

$$W_u = \text{diag}([1/\mu F_{zfl} \quad 1/\mu F_{zfr} \quad 1/\mu F_{zrl} \quad 1/\mu F_{zrr}]) \quad (72)$$

where μ is the road friction coefficient, F_{zij} is the vertical force of each tyre, $ij = fl, fr, rl, rr$.

Issue 3: Actuator constraints: Actuator constraints should be taken into account in the control allocation problem. In the actuator constraints, the friction circle limit at each wheel is as follows:

$$\sqrt{F_{ixi}^2 + F_{iyi}^2} \leq \mu F_{zi} \quad (73)$$

where $i = fl, fr, rl, rr$, T_{ti_min} and T_{ti_max} are the minimum driving torques and maximum driving torques of one another, respectively. F_{ixi} , F_{iyi} and F_{zi} are a longitudinal force, lateral force and vertical force of each wheel, respectively, and μ is the road friction coefficient.

To solve the problem mentioned above, there are two alternative methods: WLS and quadratic programming. Moreover, the WLS method is chosen in this paper.

The optimal torque vectoring algorithm can be formulated as a WLS problem as follows:

$$\begin{aligned} u(t) &= u_1(t) + u_2(t) \\ &= \arg \min_{\substack{u_- \leq u \leq u^+ \\ \text{Issue 3}}} \left(\|W_u u\|^2 + \zeta \|W_v(Bu - V)\|^2 \right) \quad (74) \end{aligned}$$

Issue 2
Issue 1

where ζ is typically chosen to be very large in order to emphasise the importance of the minimisation of the allocation error.

7 Simulink/Carsim test and analysis

To demonstrate the efficiency of the hierarchical control strategy, MATLAB/Simulink with Carsim software is applied. For one thing, the vehicle model is built in Carsim to replace the actual vehicle, and it is validated critically. For another, the vehicle control model is built and simulated in the Simulink/Carsim based on the hierarchical control strategy.

7.1 Carsim model validation

The validity of the vehicle model must be ensured before simulation. Considering the Carsim has no module for the 4WD electrical vehicle exclusively, this paper built the 4WD vehicle simulation module based on the Simulink and Carsim. First, setting the parameters in the Carsim interface including vehicle parameters, tyre parameters, suspension parameters and road parameters; second, setting the input/output channels; third,

conducting the co-simulation analysis of Simulink and Carsim. The simulation platform is shown as Fig. 9.

7.2 Simulink/Carsim analysis

To test the effectiveness of the proposed driving controller, simulation results are derived from comparing two types of driving controllers: uncontrolled strategy and WLS optimally distribution strategy. The sinusoidal steering wheel angle input is shown as Fig. 5, which is chosen to simulate consecutive lane change manoeuvres of the vehicle body. The initial longitudinal velocity is 72 km/h and the road friction coefficient is 0.8. Simulation results are shown as Fig. 10. The properties of the vehicle in this paper are shown as Table 1.

Figs. 10a–d show the simulation results of yaw rate, sideslip angle, β – w_r diagram and desired yaw moment under different control strategies. It is evident that the yaw rate and sideslip angle uncontrolled cannot track the desired ones. However, the yaw rate and sideslip angle simulated in Carsim under WLS control strategy attempt to approach the desired ones considering constraints of ‘friction circle’. Figs. 10c and d show the β – w_r diagram and desired yaw moment under WLS control.

As mentioned above, the WLS optimal control allocation algorithm can attempt to track the yaw rate and sideslip angle on the high friction coefficient conditions though there are errors in the tracking process.

8 Conclusion

In this paper, an observer is designed based on the UKF algorithm, considering the lateral tyre–road force and sideslip angle are not measured directly. Then, a hierarchical control strategy is proposed to improve the vehicle handling stability. The overloop controller is designed based on the sliding model control, and the servo-loop controller is designed based on the optimal control allocation algorithm – WLS. Finally, the control strategy is applied to the vehicle model in Simulink/Carsim to testify the efficiency of the hierarchical control strategy.

The results imply that the state observer is able to effectively estimate the lateral tyre–road force and sideslip angle no matter on high friction coefficient or low friction coefficient conditions. Moreover, the proposed control strategy is able to effectively stabilise the vehicle during sinusoidal steering angle under critical driving conditions.

As future work, we have planned to validate the proposed methods, in this paper, by hardware-in-the-loop. Additionally, we plan to implement the controller on the EV.

Besides, it is worth noting that DYC method has a great impact on longitudinal velocity though DYC could stabilise the vehicle effectively. In the near future, to improve the stability and decrease the impact on longitudinal velocity, we have planned to design an integrated controller of four-wheel-steering and DYC based on other optimisation methods. Furthermore, compared to the proposed hierarchical controller in this paper, the centralised controller could calculate the target longitudinal forces directly. For this reason, as future work, we will design a centralised controller to improve the stability of the vehicle.

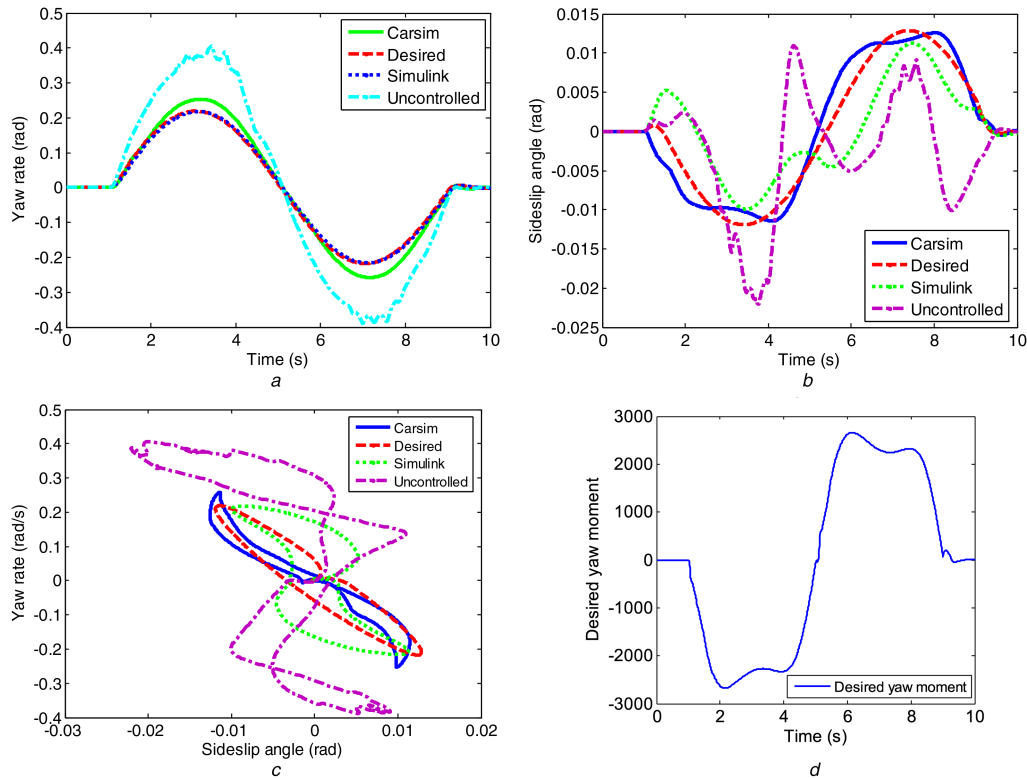


Fig. 10 Simulation results under different controls

(a) Yaw rate, (b) Sideslip angle, (c) β - $\dot{\psi}$ diagram, (d) Desired yaw moment

Table 1 Properties of vehicle

Parameters	Symbol	Unit	Values
total mass	m	kg	1416
sprung mass	m_s	kg	1274
acceleration of gravity	g	m/s^2	9.8
distance from centre of gravity (CoG) to the front wheel	a	m	1.135
distance from CoG to the rear wheel	b	m	1.555
height of CoG	h_s	m	0.652
moment of inertia about the x-axis	I_{xx}	$\text{kg} \cdot \text{m}^2$	744
moment of inertia about the z-axis	I_{zz}	$\text{kg} \cdot \text{m}^2$	3048.1
product of inertia about roll and yaw axes of the sprung mass	I_{xz}	$\text{kg} \cdot \text{m}^2$	21.09
moment of inertia of wheels	I_w	$\text{kg} \cdot \text{m}^2$	1.5
radius of wheels	R	m	0.313
track width	t_f	m	1.384

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